

Probing the Correlation Length of Metastable Annuli through Thermal Erosion in the 2D Ising Model

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In this paper, we study the stability and thermal interactions of annular structures in the 2D Ising Model, using Monte Carlo simulations. This class of structures is homotopy equivalent to rings, whose stability at constant temperature can be classified by a single parameter: its thickness W . Near a critical thickness W_c , meta-stability is strongly heightened (especially in the low temperature limit). We observe that W_c is temperature-invariant below an experimentally derived $T_c^{\text{ring}} = (0.8224 \pm 0.0041)T_c$ where T_c is Onsager’s critical temperature. At this value, the correlation length (driven by thermal fluctuations) approaches W_c , immediately collapsing any ring configuration.

I. INTRODUCTION

The 2D Ising model, despite its simple appearance, is an extraordinarily useful tool for observing the internal spin coupling of ferromagnetic materials, and their relationship with external fields. This paper investigates the convergence towards metastable states in such models (and their underlying phase transitions), motivated by magnetisation dynamics and the correlation length for states in defined topological structures. The model also provides motivation for studying skyrmion dynamics, which has applications in data storage. Our study focuses on topological stability, specifically the collapse of annular structures which is analogous to the collapse of skyrmions when pushed beyond stable limits [1].

The spin states $S = \{s_1, \dots, s_{N^2}\}$ of a 2D array (collectively, a canonical ensemble) of magnetic moments can be represented in a regular binary grid, in which each spin $s_i \in S$ (either spin-up ($s_i = 1$) or spin-down ($s_i = -1$) in a magnetic interpretation) interacts with its neighbouring states, along with external magnetic fields. These nuanced interactions have associated energies $\mathcal{J} : S \times S \rightarrow \mathbb{R}$ and $\mathfrak{h} : S \rightarrow \mathbb{R}$ respectively, giving a statistical Boltzmann probability of that cell ‘flipping’ ($s_i \rightarrow -s_i$), $P(i, S, \mathcal{J}, \mathfrak{h}) \equiv P(s_i)$. Using the nearest-neighbour construction of the model, and assuming a *uniform* magnetic field, we can simplify:

$$\mathcal{J}(s_i, s_j) = \begin{cases} 0 & i, j \text{ are not neighbours} \\ J s_i s_j & \text{else (giving } \pm J) \end{cases}, \quad \mathfrak{h}(s_i) = h,$$

where J, h are constants, and $h = gB$ [2]. From this, we can form the Hamiltonian [3] (representing the system’s configurational energy [4, 5]),

$$\mathcal{H}(S) = - \sum_i \left(\overbrace{h s_i}^{\text{external}} + \overbrace{\sum_j \mathcal{J}(s_i, s_j)}^{\text{internal}} \right), \quad (1)$$

To derive $P(s_i)$, we utilise the mean-field approximation [6, 7], which states that $\mathcal{J}(s_i, s_j) \approx m := \langle S \rangle$ [8]. In other words, the strength of neighbouring spin interactions is governed by the average degree of order prevailing in the

system [9]. Consequently, the energy associated with a particular spin state s_i ’s value becomes [4, 10]

$$\epsilon(s_i) = -s_i \left(h + J \sum_k^{\text{nbs}} \langle s_k \rangle \right) \quad (2a)$$

$$= s_i E, \text{ where } E = -(h + 4Jm). \quad (2b)$$

Thus, using the Boltzmann value Z , we evaluate [4, 11]

$$m = \frac{1}{Z} \sum_i \exp[\epsilon(s_i)/k_B T] = \tanh \left(\frac{h + 4Jm}{k_B T} \right). \quad (3)$$

For $h = 0$, this yields the trivial solution $m = 0$, and (for $T \ll$ some *critical temperature* T_c^{th}) $m \approx \pm 1$. For $T \gg T_c$, we approach asymptotic equiprobability of states through inherent random temperature fluctuations [12, 13]; as $T \rightarrow 0$, the system entropically favours local alignment of spins [14, 15], whose direction of alignment (ie: the sign of $\lim_m = \pm 1$) can be forced via the orientation of h . Using (3), one can show that $T_c^{\text{th}} = 4J$, which (in finite dimensions $d < 4$ [16, 17]) is significantly greater than those obtained experimentally [9, 18], despite recovering the correct m phase transition behaviour as compared to the high-dimensional limit [17]. The limitations of this (mean-field) theoretical approach lie in the loss of *fluctuation terms* [5, 19] between neighbouring spins (whose effects exponentially grow as $T \rightarrow T_c$ [19]), by ‘smoothing’ states $\langle s_i \rangle \rightarrow m$. Onsager proved that, for isotropic ferromagnetic systems, the critical temperature in fact adheres to [5, 20, 21]

$$\sinh^2 \left(\frac{2J}{k_B T_c} \right) = 1, \implies T_c := \frac{4J}{k_B \cosh^{-1}(3)}. \quad (4)$$

The difference $|T_c^{\text{th}} - T_c| \gg 0$ motivates the use of the stochastic Metropolis algorithm for simulating 2D Ising effects [22]. This Monte-Carlo-based approach avoids mean-field approximations, as the exact value of some neighbouring s_j can be retrieved in $O(1)$ via a 2D array structure (meaning (2a) can be calculated without the simplification in (2b), giving a more accurate value of $P(s_i)$ [22]). Utilising importance sampling and the detailed balance condition [5] in this regime results in

experimentally-derived critical temperatures consistent with that of (4) [20, 23, 24], enabling us to more effectively study the evolution of 2D Ising system [25]. We therefore quantify the dependence of nearby spins through the correlation $G(r) = \langle s_i s_j \rangle - \langle s_i \rangle \langle s_j \rangle$, where r is the distance between spins s_i and s_j . In the low temperature limit, this behaves as $G(r) \sim e^{-r/\xi}$ [15], where we define ξ as the characteristic correlation length for temperature fluctuations [26]. However, as $T \rightarrow T_c$, ξ diverges according to $\xi \sim |T - T_c|^{-\nu}$ [20, 27]. This results in arbitrarily large ‘clusters’ of aligned spins with size $\sim \xi$.

While ξ is a fundamental property of the Ising model, measuring it directly often requires complex autocorrelation features [28]. We propose a topological approach: can the stability of a metastable ring serve as a direct proxy for ξ ? By subjecting a down-spin ring to thermal erosion, we aim to investigate the ring’s collapse at different widths and temperatures in order to measure the critical thermal fluctuation length ξ .

II. THEORY

In the Ising model, temperature fluctuations lead to spontaneous creation of aligned spin domains. The behaviour of these domains is characterised by the effective forces on their boundaries.

The stability and evolution of spin domains are governed by the interactions of the Ising lattice and the external magnetic field. In this discrete system, the exchange interaction J seeks to minimise the number of anti-parallel pairs. The exchange energy, given by (1) in the case of $h = 0$, is lowered when neighbouring spins are parallel. At any interface between up-spins ($s = +1$) and down-spins ($s = -1$), every mismatched pair contributes a net energy cost $\Delta E = 2J$ per boundary bond. The line tension is the sum of these mismatches along the entire length of the interface, resulting in an edge energy of $E_{edge} = \gamma L$, where L is the length of the interface and γ is the line tension coefficient [29]. For $T > 0$, as the temperature increases, the boundary becomes rough due to thermal fluctuations [30]. This roughness increases the entropy (S) of the interface, which lowers the net cost of maintaining the boundary. Onsager provided the exact analytical solution of the 2D Ising model on a square lattice [20] to be:

$$\gamma(T) = 2J - 2k_B T \ln \left[\coth \left(\frac{J}{k_B T} \right) \right]. \quad (5)$$

In our experiment, the system seeks to minimise its total energy and is under constant force to reduce the length of the boundary. This pressure is given by

$$P_{line} = \frac{\partial E}{\partial R} = 2\pi\gamma,$$

and points in the direction that shortens the boundary [31]. The pressure is countered by the Zeeman Energy,

$\mathcal{H}_Z$, equal to (1) without interacting spin components ($\mathcal{J} = 0$). The applied negative magnetic field favours down-spins, and so for every $s_i = -1$ the system gains energy stability. The force acts to maximise the area of the state which matches the field [32]. The stability of a topology depends on if the Zeeman pressure can overcome the line pressure, which is seen in the Hamiltonian (1).

Next we consider a circular ‘nucleus’ of spins aligned along the negative axis surrounded by a sea of spins aligned along the positive axis, along with a stabilising negative external magnetic field. For simplicity we assume that on the boundary each spin will have 3 aligned neighbours (locally flat), and that entropic temperature effects are negligible. A circular nucleus with N spins will have a radius $r \propto \sqrt{N/\pi}$ and a boundary length $C = 2\pi r = 2\sqrt{\pi N}$. Using (1), we can calculate the energy cost ΔE of forming the nucleus as the sum of field and boundary energy costs [26]:

$$\Delta E(\{s_i\}, h) = 2(J\sqrt{\pi N} + hN). \quad (6)$$

At small N , the line pressure dominates. The system wants to minimise its energy, so the small nucleus is energetically unfavourable, and will eventually be dissipated by the sea of up spins. At large N , the Zeeman pressure dominates, so the nucleus will grow until it encompasses the whole domain. In between the two regimes, there is a critical radius R_c at which the two outcomes are equally likely to occur. By finding where system energy is maximal (by solving $N = N_c$), we discover:

$$\left. \frac{d(\Delta E)}{dN} \right|_{N=N_c} = 0 \implies R_c = \sqrt{N_c/\pi} = \frac{J}{2|h|}. \quad (7)$$

In practice, the ‘locally flat’ assumption is too strong [26], and the boundary will be rough due to the discrete circular shape, as well as random fluctuations due to temperature. The radius must be larger to survive thermal erosion at its boundaries [30]. Therefore, the critical radius required is greater than the value derived theoretically - in our simulations, we recover $R_c^{\text{sim}} \simeq J/1.22|h|$.

As the temperature approaches T_c , we expect to eventually cross a temperature threshold after which thermal noise dominates and no convergence happens.

The boundary of the nucleus can only interact with the sea of spins, so it is topologically stable as the nucleus is isolated. If we were to have two boundaries with independent curvatures, they could interact, allowing us to probe the effect of scale and temperature on the interaction between the two interfaces. Only with such a model can we probe length scales in a meaningful way. The ring is the natural choice for our model due to topological collapse which is a result of the fixed inner boundary condition. Our system evolves towards a state of lower free energy through curvature driven flow [33]. This principle dictates that any curved boundary between 2 phases moves with a velocity proportional to its local curvature. The

interface velocity v at any point along the ring's boundary can be described by the Allen-Cahn equation [34]

$$v = M \cdot (\gamma\kappa \pm |h|), \quad (8)$$

where $M(T)$ is a metric of the *interface mobility* of spins ‘flipping’ at a boundary [35]. For an irregular spin-down topology, regions of high curvature yield a much higher contractive force $\gamma\kappa$ (relative to the stabilising field $|h|$), resulting in the rapid retraction of locally sharp protrusions. Hence, irregular geometries tend to devolve into a circular shape [36], where the geometry enforces a constant κ at every point, equilibrating the internal pressure. The topological collapse into a ring rather than a disk or circle is a result of the fixed inner boundary condition: ‘holes’ of up-spins inside an annulus are energetically unfavourable to flip and so behave locally as a metastable nucleus, preserving the structure.

This topological evolution is driven by the difference in interface velocities, which via (8) is characterised via $|h|$ (which aims to thicken the ring), and $\{\kappa_{\text{in}}, \kappa_{\text{out}}\}$ (which aim to thin the ring): for a smooth, discretised ring, with $R_{\text{in}} \ll R_{\text{out}}$ (the inner & outer ring radii respectively) and $h = 0$, $\kappa_{\text{in}} > \kappa_{\text{out}}$, and so the line tension acts to pull the two boundaries together to minimise the total variation. Similarly, given an asymmetry in ring thickness, $\kappa_{\text{in}} - \kappa_{\text{out}}$ will differ on each side, creating a net flux of spin-flips that shifts the ring's centre of mass concentric with the up-spin core [37]. The ring is metastable upon reaching the *annular condition* [38], which states

$$\gamma \left(\frac{1}{R_{\text{out}}} - \frac{1}{R_{\text{in}}} \right) \approx 2|h|. \quad (9)$$

We construct this structure by embedding a spin-up nucleus (radius R_{in}) within a spin-down nucleus (radius R_{out}). The up-spin inner droplet faces both line pressure (5) and Zeeman pressure towards the inside [13, 20]. If the ring width $W = R_{\text{out}} - R_{\text{in}}$ is large, the inner droplet will quickly be dissipated by these forces, and we simply reduce to the single-droplet case. However as the ring width W becomes small, the two boundaries can interact. If $W \leq \xi$ is very small, the temperature fluctuations on the outer boundary are very likely to reach the inner boundary [39]. If this happens, then at this location the ring is ‘punctured’ (i.e. the sea of up spins merges with the inner drop) and the entire structure quickly collapses due to line pressure, resulting in the entire system being spin-up.

We define the critical ring width W_c to be the ring width where these two outcomes are equally likely. This should happen near the correlation length $W_c \simeq \xi$. By varying W , we can determine the precise scale at which the temperature fluctuations bridge the gap between the sea and the inner droplet, triggering a collapse that is not driven by the energy pressures, but by the correlation between the interfaces.

As $T \rightarrow T_c \implies \xi \rightarrow \infty$, we expect that at some temperature $T_c^{\text{ring}} \leq T_c$ the correlation distance will be

much greater than the length scale of our simulation, and we will no longer be able to get meaningful results for W_c .

III. METHODOLOGY

As motivated by Section I, we make use of the Monte-Carlo based Metropolis algorithm to simulate Ising effects, for a square grid of width $N = 50$ spins. Preliminary tests showed that this is beyond the point of diminishing returns, and periodic boundary conditions are used to increase the accuracy of the results for N fixed [8]. Therefore, as we are more interested in the dependencies and linearity of W_c (rather than the specific values it takes), we follow the theory in (7), (II) and explore $h = -0.1g\mu_B$ fixed (being slightly aligned with the ring's downward spin state). This gives, for a single nucleus, $R_c^{\text{sim}} \approx 8.2$, which we expect to be on the same order of magnitude as W_c . All quantities are made dimensionless by setting $J = k_B = \mu = \nu = 1$, and with random generator (RNG) seed set to 1 for reproducibility. Defining a unit of ‘Monte Carlo Time’ (MCT) to be that of a single simulation step (in which a single spin state is considered), the processing (in Python) has a rough translation of $\sim 3.5 \cdot 10^8$ MCTs per second.

To determine when it is safe to terminate a simulation for results probing, we must classify when a system (treated as a canonical ensemble) has reached an equilibrium state. To allow the system to explore all possible states, we must perform at least an order of magnitude of $\sim N^2$ MCTs [5] (our simulation uses between $2^{10}N^2$ and $2^{16}N^2$ MCTs for each simulation). As the *equilibrium time* τ_{eq} is a good measure of the stability of initial states, it is also a good metric to quantify. As the state of $m = \pm 1$ is clearly at equilibrium, we can track m for a given state, and terminate the simulation when we surpass some non-zero bounds about ± 1 (weighted by $m(t=0)$). For $T > 0$, these bounds must be non-zero, due to inherent thermal fluctuations preventing the fully aligned system from being a stable state. Due to the stochastic nature of the simulation, it is common for two simulations in the same initial state to take different paths to equilibrium [5]. To avoid this potential pitfall, we further typically run multiple simulations for the same initial state in parallel, then define $t = \tau_{eq}$ as the point where the two systems reach approximately the same, *unchanging* value (of some quantity of interest) [5].

IV. RESULTS

Exploring the metastable 50% region in Figure 1 (ie: the cells fitted to W_c) shows heightened ring ‘stability’ (illustrated also by the greater time of convergence). We see that, for $R_{\text{in}} \lesssim 5.5$, the system's behaviour is identical to a single-drop nucleus of radius R_{out} : the droplet's spins do not ‘see’ the negligible ring structure. We observe this behaviour until R_{in} increases to $R_{\text{out}} - W_c$, at which the

structure behaves as a stable ring. We also see that, for $R_{\text{out}} > R_c$, W_c remains at a constant (2.9571 ± 0.027) cells: W_c is not a function of the ring's dimensions.

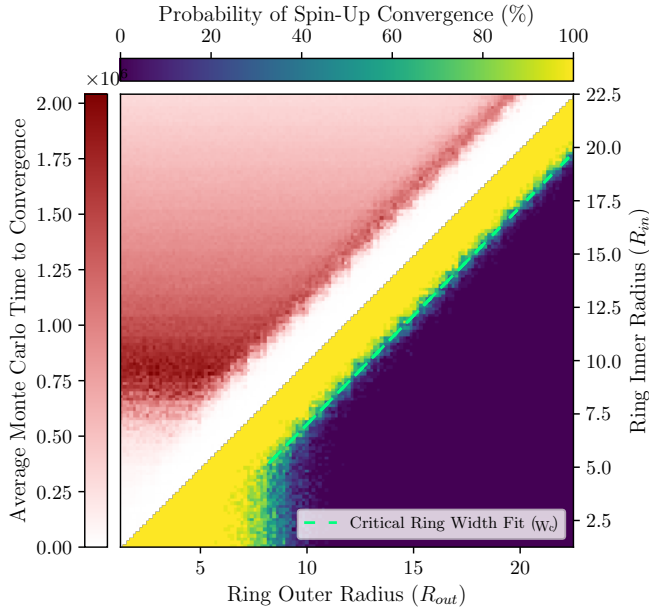


FIG. 1. A parameter exploration of possible initial ring configurations, with temperature $T = 0.5T_c$. For each parameter set, we average the convergent behaviour of 40 identical states, and use this to estimate the probability of the ring collapsing into a spin-up state (yellow) or expanding into a spin-down state (purple). The upper triangle is a reflection of the lower, and gives an illustrative measure of how long it takes each initial ring state to converge (on average).

Inspired by this, we explore the temperature dependence of W_c in Figure 2 by fixing $R_{\text{out}} = 15$ (a suitable size, such that the sea of up states encompassing it is $\gg W_c$). For each temperature, we sweep 30 values of $R_{\text{in}} \in [0, R_{\text{out}}]$, to determine at which radii we get stable ring structures (represented by a stark shift from down-spin convergence to up-spin convergence). If this crossing behaviour does not emerge for a specific configuration, *no* initial ring configuration can be metastable, and thus we set $W_c = 0$.

The results show that W_c behaves like a step function, due to the divergence of the correlation length ξ . In the low temperature regime, the ring's width W is significantly greater than the scale of thermal fluctuations $\xi \ll W_c$, effectively isolating the inner and outer interfaces of the annulus. As $T \rightarrow T_c^{\text{ring}}$, ξ increases until a critical threshold relative to W , at which the thermal noise bridges the gap between the ring's boundaries. This creates a path for the up-spin phase to penetrate the down-spin, leading to loss of topological stability. Beyond this temperature, the ring is no longer metastable; the line tension now dominates the structure, leading to the immediate collapse we observed.

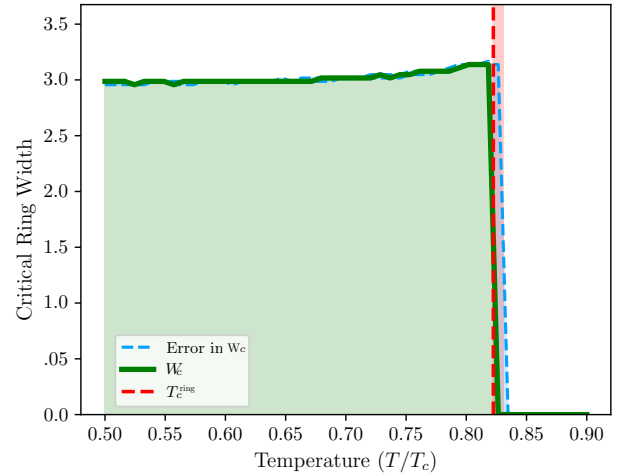


FIG. 2. A study of the temperature dependence of W_c , for $T \in [0.5, 0.9]T_c$ (plotted in green), with error ranges (barely noticeable, due to averaging over 150 simulations per batch) in blue. We observe a remarkably constant behaviour of $W_c \approx 3.0$ cells, until we reach a temperature of $T_c^{\text{ring}} := (0.8224 \pm 0.0041)T_c$. Above this temperature, W_c immediately drops to 0. Due to the wringing instabilities about $\{T_c^{\text{ring}}, W_c\}$, a Gaussian distribution (centred about these values) is applied.

V. CONCLUSION

In this paper, we investigated the meta-stability of various annuli in the 2D Ising model, which we demonstrate are homotopy equivalent to the ring structure by considering the effects of line pressure. By investigating the thermal erosion of such systems using the Monte-Carlo Metropolis algorithm, we successfully identified a distinct phase transition about $\xi \approx W_c$. We hypothesise that this is supported by the marginal increase in W_c in Figure 2, as $T \approx T_c^{\text{ring}}$. In this region, thermal fluctuations on the inner and outer interfaces overlap, allowing the up-spin phase to ‘puncture’ the annulus and trigger an immediate topological collapse. Otherwise, W_c exhibits invariance to temperature and ring parameters.

While the discretisation of our finite grid limited the resolution of this rising effect, the sharp transition to $W_c = 0$ clearly marks the point where thermal noise consistently overwhelms the energetic barriers favouring spin alignment. These findings should motivate future research to rescale these simulations, allowing for higher resolution investigation of ring width variances in the narrow neighbourhood about T_c^{ring} . While the Metropolis algorithm is a robust standard, the inherent ‘stiffness’ of the discrete Ising lattice may have obscured the finer dynamics of the interface [30]. The Ginzburg-Landau model (in which spin states are represented as continuous variables) [34] could more accurately display the difference between the interfacial tension and the thermal noise, providing a better view of how correlation lengths drive topological failure in metastable structures.

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